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Vellore Institute of Technology
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CSI4901 – Capstone Project

Fall Semester 2024-2025

Zeroth Review

HOSPITAL MANAGEMENT SYSTEM USING MERN STACK

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ONE PAGE WRITEUP

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Project Title :- Hospital Management System Using MERN Stack.

Introduction:

A hospital management system is a comprehensive software solution designed to streamline various administrative, clinical and financial processes within a healthcare facility. It is designed to manage various functionalities such as booking an appointment, doctor giving the prescription. By integrating with MERN stack technology into HMS, we can reduce the paper work and improve efficiency of doctor-patient interactions.

When built using MERN Stack (MongoDB, Express.js, React.js, Node.js) the system develops a robust, scalable and flexible architecture. MongoDB serves as database to manage large volumes of patient data, medical records, appointments. Express.js and Node.js for backend framework. React.js for frontend which can provide a user-friendly interface for both users as well as admins.

Objectives:

- * To automate and simplify administrative tasks such as patient registration, appointment scheduling, billing and inventory management.
- * To reduce manual effort, minimize errors and improve the operational efficiency.
- * To develop an intuitive and responsive user interface that caters to the needs of various users, including doctors, nurses, staff and patients.
- * To design a scalable system that can accommodate the growing needs of the healthcare facility, where addition of new features or expansion of website.

Technologies :

- 1) MongoDB : Serves as database to store user credentials as well as appointment booking.
- 2) Express.js, Node.js : Handles the backend logic to process patient data from the frontend and passes into the database.
- 3) React.js, Bootstrap, CSS : Develop the frontend logic to create a visually appealing and user-friendly interface.
- 4) Dotenv : A module that enables the use of environment variables in a Node.js application. Dotenv is used to manage all our sensitive configuration variables, such as database connection strings, API port number and secret tokens.

Review - O Write up

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BASE PAPER

Wang, M., Le, Y., & Yu, C. H. (2023, December). MedCare4Home: A Mobile Medical System for Families. In *2023 IEEE International Conference on E-health Networking, Application & Services (Healthcom)* (pp. 259-264). IEEE.

TIMELINE CHART

